

Damien MAÎTRE

MSc Engineering Student — CFD, HPC & Numerical Methods

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EDUCATION

September 2026 February 2027	MSc Mobility & Physics engineering <i>Student exchange at Politecnico di Milano, Italy</i> International economy, transport operation, high performance computing, infrastructure finance.
September 2023 June 2027	MSc Engineering — Fluid dynamics, Structural mechanics, Prototyping <i>École nationale des ponts et chaussées (ENPC), Paris</i> Structural mechanics and large-structure design, FEM applied to structural analysis, composite materials, metallurgy, complete fluid dynamics module, HPC, combustion. • Tools: OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, gmsh • Projects: Underwater sea-glider simulation and prototype ; closed-wing aircraft design and optimization ; wooden suspension bridge design, verification and construction.
September 2021 June 2023	Bachelor — Classes préparatoires <i>Lycée Kléber, Strasbourg</i> Intensive programme in mathematics, physics and engineering sciences.

RESEARCH PROJECTS

Complete CFD developer-first suite

Personal project — OpenFOAM, Gmsh (forked), Python, C++

Open and user-friendly aerodynamical loads studies, tailored for very high Re axisymmetric problems.

Structural FEM & Eurocode Verification suite

Personal project — OpenSeesPy, Python

End to end structure design, visualization and verification tools.

PROFESSIONAL EXPERIENCE

February 2026 July 2026	Rail infrastructure engineering internship — TRACTEBEL FRANCE <i>CDG Express project, Saint-Ouen</i> • Construction compliance verification for the CDG Express, a 2.5B€ high-speed rail project, • Structural analysis and Eurocode verifications of civil engineering structures, • Review of structural calculation reports and as-built site surveys, • Drafting technical reports and recommendations for the client.
July 2025 January 2026	Digital Twin PM intern — SNCF Voyageurs <i>Industry & digital solutions, Saint-Denis</i> • Management of a 150k€ digital project within the rolling stock maintenance industry, • Coordination of 15 3D artists across the country's maintenance plants, • Project ownership and reporting to internal clients (plant directors, Industrial Director, etc.), • End-to-end procurement and tendering process.

TECHNICAL SKILLS

CFD & FEM	OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, Gmsh
Programming	10+ years of experience with Python, C++ and UNIX environments

LANGUAGES

French — Native · **English** — Proficient (C1) · **Italian** — Working proficiency (B2) · **German** — Working proficiency (B2)

ACTIVITIES & COMMUNITY INVOLVEMENT

Rescuer — French Red Cross

Volunteer first-aid responder

First-aid responder at public events (marathons, festivals, Paris-Plage) and on-call with Paris Fire Brigade.

Director — Student Musical Ensemble

BridgeWay Comédie Musicale, Champs-Sur-Marne

Directed an ensemble of 75 artists and a creative/technical leadership team of 12 with 4.5k€ budget.

General Secretary — Student Association

Union des Elèves de l'Ecole Nationale des Ponts et Chaussées

Management of a team of 15 for professional and social events with 1,000+ attendees, Administrative management of a registered French non-profit (Loi 1901).